<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U03E615</td><td colspan="5">Cloud Computing</td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td colspan="2">P</td><td>TUT</td><td>Total</td></tr>
<tr><td>03</td><td colspan="2">--</td><td>--</td><td>03</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td colspan="2">--</td><td>--</td><td>03</td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>--</td><td>20</td><td>30</td><td>50</td><td>100</td></tr>
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Course pre-requisites: Computer communication networks
Network security

Course Objectives: Cloud computing services are being adopted widely across a variety of organizations and in many domains. This course will introduce the fundamental concept of cloud computing, it's enabling technologies, virtualization and security concerns. This course also imparts the functional capabilities of popular cloud platforms such as GAE, AWS and Azure.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand different cloud service models
CO2. Differentiate between various virtualization techniques
CO3. Recognize different cloud programming structures and usage from business perspective
CO4. Understand several security mechanisms for cloud environment
CO5. Understand challenges of maintaining security in cloud architecture

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	**Cloud Computing fundamentals**		**08**	**CO1**
	1.1	Overview of Distributed Computing, Cluster computing, Grid computing.		
	1.2	Evolution, characteristics, and components of cloud computing, Advantages of Cloud computing		
	1.3	Cloud service Models, Cloud Deployment Models , Alternative Deployment Models		
	1.4	Relevant Deployment Models for Enterprise Cloud Computing		
	1.5	Centralization of Data Storage		
2	**Virtualization techniques**		**10**	**CO2**
	2.1	Virtual Machines and Virtualization of Clusters and Data Centers, Virtualization Structures/Tools and Mechanisms		
	2.2	Implementation Levels of Virtualization, Server Virtualization: Understanding Server Virtualization, types of server virtualization		
	2.3	Virtualization Structures/Tools and Mechanisms, Virtualization of CPU, Memory and I/O Devices		
	2.4	Virtual Clusters and Resource Management, Virtualization of Data-Center Automation		
3	**Cloud computing architectures and Data Centers**		**09**	**CO3**
	3.1	Data-Center design and Interconnection networks		
	3.2	Architectural Design of Compute and Storage Clouds		
	3.3	Virtual machine migration and Load balancing		
	3.4	Web services - Web 2.0 - Web OS - Case studies – Anything as a service (XaaS).		
4	**Cloud Platforms: overview and services**		**10**	**CO4**
	4.1	**Programming support of Google App Engine:** Programming the Google App Engine, Google file system(GFS), Bigtable, Googles NOSQL system. Chubby, Google's Distributed lock service		
	4.2	**Programming on Amazons AWS:** Programming on Amazon EC2, Amazon simple storage service(S3), Amazon Elastic block store(EBS), and SimpleDB, Virtual private cloud		
	4.3	Programming on Microsoft Azure: Core architecture components of Azure, Azure compute, networking, storage services		

5	Security architecture and Challenges		08	CO5
	5.1	Architectural Considerations- General Issues, Trusted Cloud Computing, Secure Execution Environments and Communications, Micro-architectures; Identity Management and Access control-Identity management, Access control		
	5.2	Virtualization Security, VPN, Management Virtual Threats , Hypervisor Risks ,Increased Denial, of Service Risk , VM Security Recommendations		
	5.3	Cloud federation,: EU cyber solidarity act, interoperability and standards: ISO 27018		
		Total	**45**	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	KaiHwang, Geoffrey C. Fox, Jack J. Dongarra,	*Distributed And Cloud Computing From parallel processing to the internet of things*	ELSEVIE MK publishers	First, 2012
2	Ronald L. Krutz, Russell Dean Vines	*Cloud Security A Comprehensive Guide to Secure Cloud Computing*	Wily	First, 2010
3	Antohy T Velte	*Cloud Computing : A Practical Approach*	McGraw Hill	10th Edition 2004
4	Rajkumar Buyya, James Broberg, Andrzej Goscinski	*Cloud Computing, Principles and Paradigms*	Wiley	First ,2013